Campus Item Exchange

Phases 4 and 5 Architecture and Detailed Design | 30 October 2026 | Academic prototype

Purpose and scope

An internal campus system for finding, requesting, transferring, maintaining and tracking reusable assets. The prototype covers one campus, two sample departments and about 20 labelled items. AI supports people with advice; permissions, stock, approvals and calculations remain controlled by application rules.

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Review and individual contributions

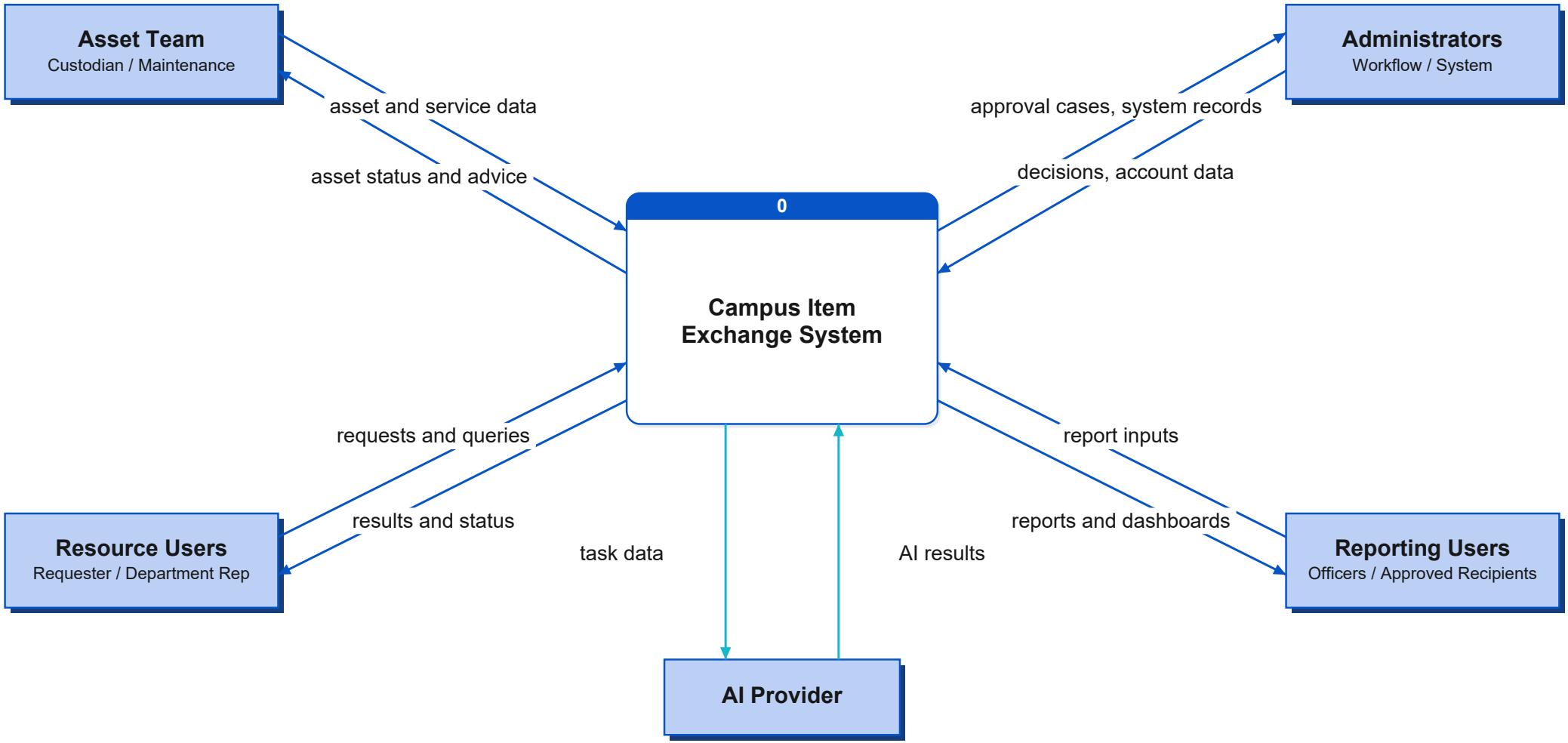
The four members and IDs match Phases 1-3. Each student must review and revise their contribution and commit it using their own identity. This document does not assert that individual GitHub commits or implementation tests have been completed.

Requirements and document status

Phase 3 requirement IDs appear in every section. Example records, AI outputs and screen values are synthetic. Deadlines, review gates and acceptance checks are proposed design settings; measured implementation results belong in later test records. This covers Phases 4 and 5.

Baseline: Campus Item Exchange Phase Three Requirements Document, 4 October 2026; Phase 4 Architecture Design lecture, Fall 2026.

Campus Item Exchange context diagram

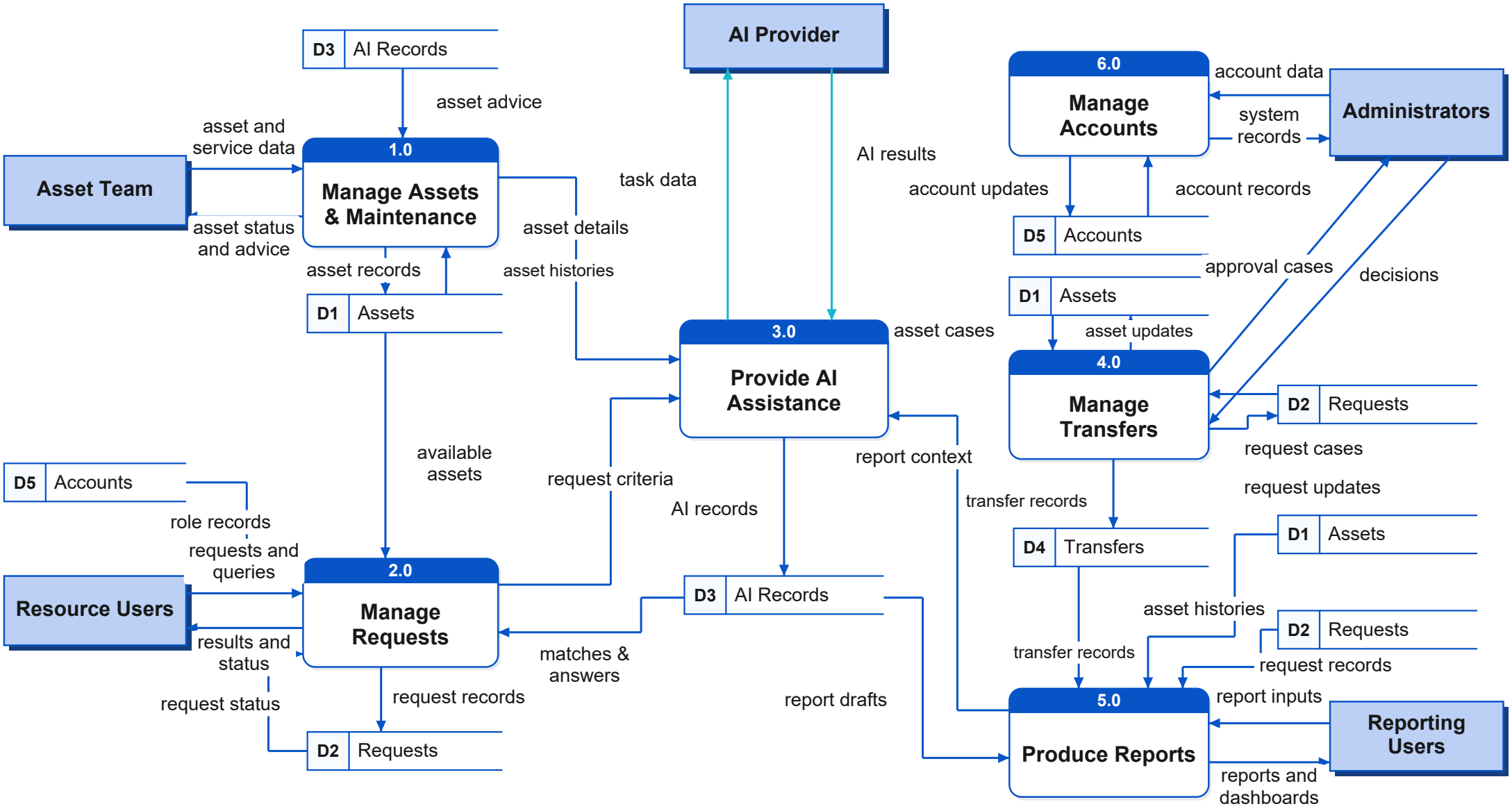


One process (0), grouped noun flows, no internal steps or data stores.

Reporting officers are Procurement, Finance and Sustainability. Grouping roles does not combine their permissions.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

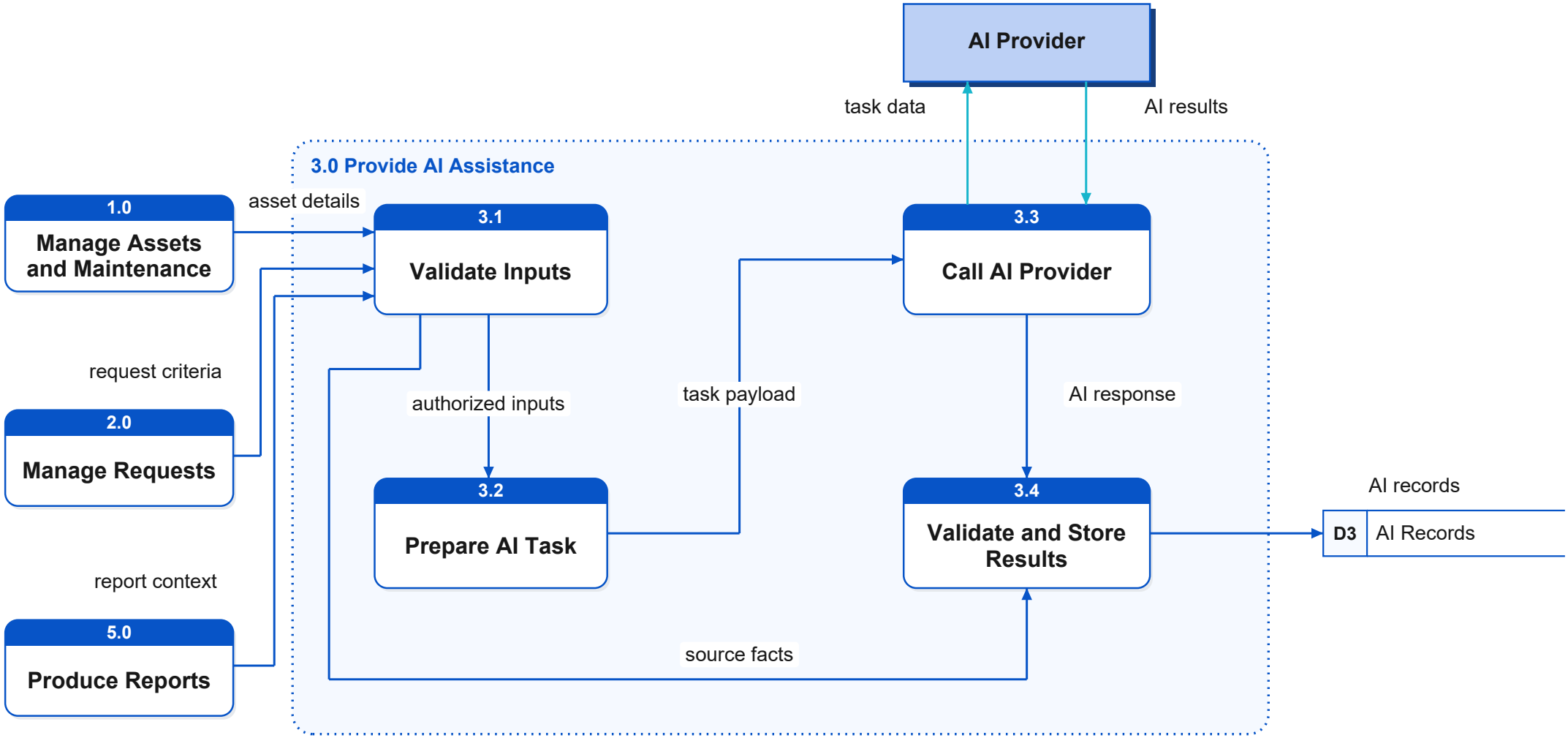
Campus Item Exchange Level 1 DFD



Repeated D-numbers refer to the same store. Maintenance is included in 1.0; approvals and disposition are included in 4.0.
Role families cover the full project. Sign-in, audit and notifications remain shared controls; permissions stay separate.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

Campus Item Exchange Level 2 of 3.0



The six flows crossing the dotted boundary are exactly the six flows of 3.0 on Level 1.

3.0 is expanded because it handles untrusted AI responses. Validated results or failure status are stored before returning to callers.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

Campus Item Exchange Level 1 balancing table

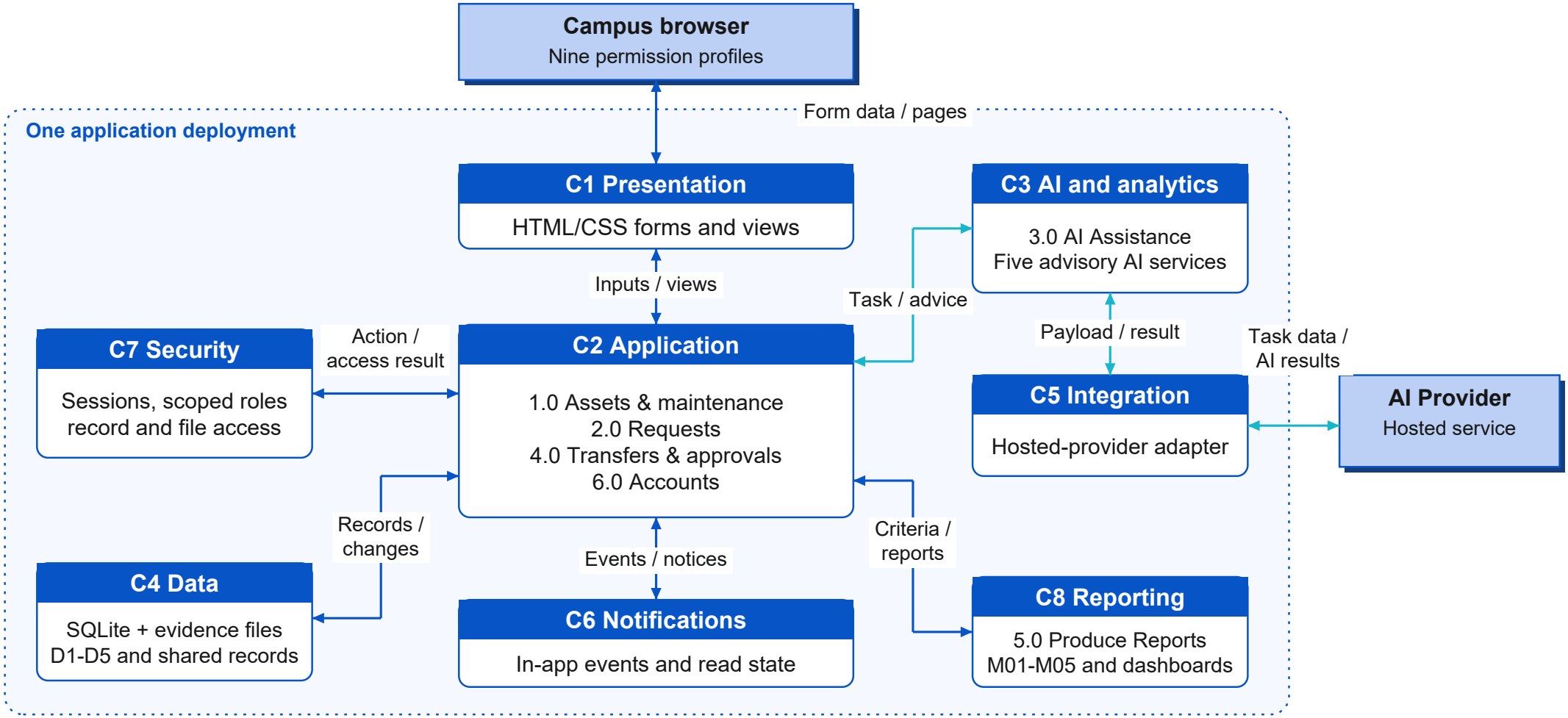
#	Context flow	External entity	In/out	Level 1 process	OK
1	asset and service data	Asset Team	in	1.0	✓
2	asset status and advice	Asset Team	out	1.0	✓
3	requests and queries	Resource Users	in	2.0	✓
4	results and status	Resource Users	out	2.0	✓
5	decisions, account data	Administrators	in	4.0, 6.0 (split)	✓
6	approval cases, system records	Administrators	out	4.0, 6.0 (split)	✓
7	report inputs	Reporting Users	in	5.0	✓
8	reports and dashboards	Reporting Users	out	5.0	✓
9	task data	AI Provider	out	3.0	✓
10	AI results	AI Provider	in	3.0	✓

VERIFY 10 of 10 match: same name, same direction, same entity. Grouped flows (5, 6) split at Level 1.

In/out is relative to the whole system. Flows 5 and 6 split between 4.0 Manage Transfers and 6.0 Manage Accounts.

3.1 Campus Item Exchange architecture view

Modular monolith: one Flask application, a private data environment and one external AI provider.



Routes call services; services use scoped repositories and coordinate AI and reports. Security applies to every server component.

Manual search, requests, asset history and numeric reports remain available during an AI failure. Paired labels show exchanged data.

Drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR09-NFR11; BR01-BR07.

3.2 Architecture decision records

Each record compares two options, explains the choice and trade-off, and states a planned verification check.

ADR-01 Architecture style

Drivers: FR01-FR18; NFR05, NFR12; BR01-BR07.

Option A
Layered monolith: one application organized by presentation, business logic and persistence.

Option B
Modular monolith: one application organized around business responsibilities, with shared services.

Choice We choose a modular monolith because six DFD responsibilities give the team clear module boundaries while reservations and transfers can share local transactions.

Give up We give up the simpler layer-only organization. Module boundaries need discipline; deployment, restarts and scaling remain shared.

Check Trace 1.0-6.0 to their owning components. Check module interfaces and test reservation and transfer rollback without partial changes.

ADR-02 Presentation and application stack

Drivers: Phase 3 Section 1 baseline; FR11, FR16; NFR02-NFR04, NFR07-NFR08, NFR12.

Option A
Flask with server-rendered HTML/CSS, ordinary forms and limited browser scripting where needed.

Option B
Flask JSON API plus a separate React browser application, with coordinated client and API behavior.

Choice We choose Flask with HTML/CSS because it preserves the Phase 3 baseline and keeps the four-person prototype manageable. All forms call the same authorized application services.

Give up We give up a separately maintained client application and some browser interaction flexibility; additional clients would need API endpoints.

Check Test login, request and approval flows in the recorded Chrome and Firefox versions at 1280 x 720, including keyboard use and forbidden direct requests.

3.3 Data and AI decision records

The technology choices preserve the Phase 3 prototype baseline. Provider capabilities still require confirmation.

ADR-03 Prototype data and evidence storage

Drivers: Phase 3 Section 1 baseline; FR05, FR08, FR12-FR13; NFR05, NFR09, NFR15.

Option A
SQLite and protected evidence files, with scoped repositories, explicit transactions and coordinated snapshots.

Option B
PostgreSQL and protected evidence storage, with a separate database service to configure and recover.

Choice We choose SQLite because the baseline is one campus, two sample departments and about 20 items. Transactions must recheck availability and prevent duplicate reservations or handovers.

Give up We give up an independently deployed database service. The prototype must pass the required concurrent-write and growth checks before acceptance.

Check Race requests for the last unit; retry a handover; restore database and evidence within 10 minutes. Run NFR15 with 200 items and 10 sessions.

ADR-04 AI gateway and provider integration

Drivers: AI01-AI08; NFR03-NFR04, NFR06, NFR10, NFR14; FR05, FR17.

Option A
One internal AI gateway and provider adapter: classification, semantic matching, ranking, action advice and LLM assistance/report drafts.

Option B
Direct provider calls in each feature module, with separate validation, privacy, timeout and fallback handling.

Choice We choose one gateway because shared task contracts keep authorized payloads, output validation and provenance consistent. AI suggests; application rules and human approvals control changes.

Give up We give up direct feature-specific provider access and accept a common dependency for all AI features. Contract changes require coordination.

Check Test all five services with success, timeout and invalid output. Enforce 15 seconds per AI call; preserve manual workflows and model/review records.

AI-use statement: AI assisted with drafting, figures and consistency checks. Team review and implementation testing are pending.

4.1 AI service plan

A = ask and wait. All services run through 3.0 Provide AI Assistance and the shared provider adapter.

Service	Pattern	Why	Time limit	If down or invalid	If unsure
Asset classification AI02	A	Custodian is editing one asset and can continue manually.	8 s	Preserve draft; select category and tags manually. Explain unsupported evidence.	Below 0.80 or confidence Unavailable: manual-review warning. Never pre-accept; unknown facts stay Unknown.
Semantic matching AI01	A	Small dataset; compare current authorized candidates without a maintained vector index.	At most 6 s; within shared 12 s	Keep need and filters; show labelled keyword/rule results. Skip AI ranking after matching failure.	After hard checks, compatibility below 75/100 means weak match. Unavailable confidence needs review; no default choice.
Ranking and explanations AI06	A	User needs one ordered list from the same eligible candidate set.	At most min(6 s, shared time left)	Keep candidates in labelled rule order; show known factors without invented AI scores.	No top-pick emphasis if top score <75, top-two gap <10, required fact missing or confidence <0.80/Unavailable.
Sustainable-action advice AI03	A	User asks about one asset; its ordinary service/disposition form remains available.	12 s	Preserve evidence and use normal maintenance/disposition forms; old advice shows its date and stale warning.	Below 0.80, Unavailable confidence or missing evidence: qualify advice. No invented saving or safety clearance; human approval stays required.
LLM assistant and report drafts AI04-AI05	A	Person requests an answer or optional draft; numeric reports appear independently first.	12 s per explicit answer or separate draft	Retain question; offer normal screens. Keep numeric report visible and label the narrative unavailable.	Cite sources or say unavailable. Reject false IDs, policy or totals. <0.80/Unavailable requires review; requests remain previews until confirmed.

Budgets include retrieval, provider transport and validation. The Phase 3 limit is 15 seconds per AI call.

Drivers: AI01-AI08; FR01-FR05, FR09-FR10, FR13, FR18; NFR03, NFR06, NFR10, NFR13-NFR14; UC09-UC13.

4.2 Shared AI controls and verification

Thresholds are proposed prototype review gates. Confidence and compatibility have different meanings.

Deadline

Every budget includes scoped retrieval, provider transport and validation. Matching and ranking share one 12-second deadline: matching <=6 seconds, ranking <=min(6, remaining). No fresh deadline starts between stages. NFR06 remains 15 seconds per AI call.

Failure and capacity

No automatic retry or unbounded waiting. Return AI busy promptly if no configured capacity is available. Reserve web capacity for ordinary screens. A new attempt needs an explicit action; late results cannot replace a timeout or change records.

Confidence

Use a numeric confidence only if its origin and 0-1 scale are documented for that task. The proposed warning threshold is 0.80. Otherwise show Unavailable and require manual review; never invent confidence or treat an LLM's self-reported number as calibrated accuracy.

Compatibility

The proposed normal-match threshold is 75/100. Top-pick emphasis also needs a 10-point gap when two candidates exist, complete required compatibility facts and no unresolved confidence warning. These are display/review gates, not accuracy claims or permissions.

Validation and authority

Check current identity, record scope, source revision, response schema, IDs, categories, facts and report totals. Recheck stock and safety when a user confirms. No score can override a failed rule, authorize a disposition or clear an unsafe asset.

Provenance

Store input references/snapshots, time, provider, model/version or Version unavailable, prompt version, output, confidence or Unavailable, validation outcome, and accepted/edited/rejected/overridden decision with reason. Exclude passwords, session secrets and AI keys.

Planned acceptance checks

Demonstrate successful results for all five services, not only fallback screens. Test at least 8/10 suitable top-three matches and 8/10 correct labelled classifications, including supported image/PDF input.
Check ten explanations against real factors; test department-name swaps, fabricated IDs, wrong totals, revoked roles, stale inputs and late/duplicate responses. Record tester, environment, date, expected/actual outcome and evidence.
Before acceptance, confirm provider/model versions, evidence handling, documented confidence semantics and actual capacity. Numeric reports remain available before a separately requested narrative draft.

AI07-AI08; FR12, FR16-FR17; NFR03-NFR06, NFR10, NFR13-NFR14; BR01, BR04-BR07

4.3 AI pattern decision records

A = ask and wait; B = compute ahead; C = queue and notify. Each service has a real alternative and a planned check.

Record / service	Two options	Choice and because	What we give up	Planned check
ADR-05 Classification	A: interactive call; B: recompute after each asset revision.	We choose A because the custodian reviews fields while editing.	We give up ready-made suggestions and completion after leaving the editor.	Test valid text/image/PDF, unsupported evidence, timeout and retained draft.
ADR-06 Semantic matching	A: compare on demand; B: maintain versioned representations on save.	We choose A because about 20 items do not yet justify index and invalidation handling.	We give up instant reuse of precomputed vectors; revisit if growth/latency checks fail.	Use ten known needs; test permission changes, stale stock and matching timeout.
ADR-07 Ranking	A: rank the current list; C: queue ranking and notify later.	We choose A because the requester is waiting to choose from one current eligible list.	We give up deferred results after the user leaves and accept a bounded wait.	Test top-two ties, unknown factors, department swaps and the shared deadline.
ADR-08 Sustainable-action advice	A: explicit foreground advice; C: queue and notify when ready.	We choose A because one-asset reviews fit the prototype and manual forms provide continuity.	We give up advice that completes after navigation away and accept a short wait.	Test missing evidence, timeout, stale advice and unchanged approval/safety controls.
ADR-09 Assistant and report drafts	A: separate requested answer/draft; C: queue longer generation and notify.	We choose A because it keeps one interaction path while numeric reports remain immediately usable.	We give up deferred completion and hidden long-running generation.	Test cited answers, exact request preview, mismatched totals and draft failure after report display.

Drivers: AI01-AI08; NFR01, NFR03, NFR05-NFR06, NFR10-NFR11, NFR14-NFR15; UC09-UC13. All settings require validation.

5.1 Human review and approval points

AI may suggest, rank, explain or draft. A permitted person decides; server checks remain mandatory.

Point	Who decides	What they review and confirm	Why / required record
H1 Metadata	Assigned asset custodian	Review entered fields, evidence, suggested category/tags, confidence and missing facts. Accept, edit or reject; confirm final fields before saving.	Risk: invented or incorrect metadata. Record input revision, reviewed values, AI provenance and edit/override reason. FR01-FR02; AI02, AI07-AI08.
H2 Request or chat preview	Requester / department representative, within scope	Review exact asset, quantity, destination, date, need and stock. Explicitly confirm: normal rules create an atomic hold. Choosing a match alone reserves nothing; unmatched needs may stay Seeking.	Risk: stale or unauthorized allocation. Record exact preview, creator, stock check, resulting request/hold or failure and AI review. FR03-FR05; AI01, AI04, AI06-AI08; BR01-BR02.
H3 Publication	Workflow administrator; no self-approval	Review release proposal, owner, condition, serviceability and evidence. Approve or reject with a reason; only approved, serviceable active units may be published.	Risk: publishing unsafe or unreleased stock. Record actor, time, reason and old/new publication state. FR02, FR17; BR01, BR04; NFR13.
H4 Reservation / transfer decision	Workflow administrator; no self-approval	Review purpose, destination, quantity, hold/expiry and current eligibility. Approval retains the same hold and changes no ownership. Rejection releases stock once.	Risk: unsupported authority or double deduction. Record case/hold IDs, actor, time, reason and status transition. FR05, FR07, FR17; BR02-BR04; NFR05.

Every gate records actor, time, case ID, reason and the previous/new state. No one approves their own request.

Drivers: FR01-FR08, FR12-FR13, FR16-FR17; AI01-AI02, AI04, AI06-AI08; BR01-BR04; NFR03, NFR05, NFR13.

5.2 Physical actions, reporting and accounts

Human judgment and evidence control changes that AI cannot authorize or verify.

Point	Who decides	What they review and confirm	Why / required record
H5 Collection and receipt	Source and receiving custodians	Source records collection arrangements and evidence. Receiver confirms the full approved receipt. Only then does one transaction update ownership/custody. Refused or overdue receipt enters dispute; units stay blocked.	Risk: unproven physical handover. Record both actors, times, evidence, quantity/condition and old/new custody, owner, location and hold. FR08, FR13, FR17; BR02-BR06.
H6 Repair / safety	Custodian authorizes repair; assigned maintenance staff records inspection	Review defect, job state, costs or Unknown and AI advice. Follow service states and record human inspection clearance before restoring serviceability. AI cannot clear an asset.	Risk: unsafe units returning to use. Record authorization, inspection evidence, service actions, condition and advice override reason. FR09, FR14; AI03; NFR13; UC07, UC11.
H7 Disposition	Custodian proposes/executes; workflow administrator approves	Review unallocated units, holds, condition, costs and evidence. Approve/reject with a reason. Retirement blocks use; final disposition requires execution evidence. No self-approval.	Risk: irreversible loss or disposal of held stock. Record proposal, review, approval, actors/times, execution evidence and lifecycle transition. FR10, FR13, FR17; BR04-BR07.
H8 Reporting	Procurement, finance or sustainability officer; workflow admin for operations	Validate only assigned inputs. Review period, record links, calculated totals, coverage and optional AI draft. Reject mismatched figures; numeric reports remain usable without AI.	Risk: invented impact or wrong period. Record input changes, contributing IDs, calculation basis and narrative review. No additional report-approval state is introduced. FR12, FR18; AI05; BR07; NFR11.
H9 Accounts / roles	System administrator	Review identity, active status, profile, department and custody. Confirm changes; reassign custody before deactivation. This profile alone cannot approve business cases.	Risk: excess access or orphaned custody. Record target account, actor, time, reason and old/new role, department and active state. Exclude secrets. FR11-FR12, FR14, FR16; BR06.

Workflow administrators resolve receipt disputes: acknowledged receipt completes; evidenced physical return permits cancellation and stock restoration.

Drivers: FR08-FR18; AI03, AI05, AI07-AI08; BR02-BR07; NFR03-NFR05, NFR11, NFR13; UC06-UC08, UC11, UC13.

5.3 Human review design decision

The interaction changes with the evidence and authority needed at each gate.

Record ADR-10	Stage-specific human review
Drivers	FR07-FR08, FR10, FR17; AI07-AI08; BR04-BR06; NFR13.
Option A	One generic human decision form for all approval gates, with case-specific validation.
Option B	Stage-specific review forms for metadata, requests, approvals, receipt and disposition, using shared authorization and audit services.
Choice and because	We choose stage-specific review forms because each gate needs different evidence, role checks and consequences. Explicit confirmation invokes the ordinary transaction; AI never applies an action.
What we give up	We give up one reusable generic screen and accept more form states and validation paths.
Planned check	Try a self-approval, an AI-only submission, a receipt without evidence, an unsafe unit and a duplicate confirmation; verify the required gates and atomic effects.

Failure and concurrency behavior

If permission, state, safety, stock or evidence changes before confirmation, reject the action and preserve valid input. Failed writes leave no partial update; repeated confirmation returns the existing result. A record correction appends a linked entry rather than erasing history.

Drivers: FR05-FR08, FR10, FR12-FR13, FR16-FR17; AI07-AI08; BR04-BR06; NFR05, NFR13.

6.1 Access matrix - requests and assets

Default: deny. A scope entry still requires an active account, permitted record, valid action and allowed state.

Action	RQ	DR	AC	MS	WA	PO	FO	SO	SA	AI
Sign in; read own notifications	Self	Self	Self	Self	Self	Self	Self	Self	Self	-
Search permitted catalogue listings	Scope	Scope	Scope	Scope	Scope	Scope	Scope	Scope	Scope	Suggest
Create/view/cancel needs and requests	Own	Dept	-	-	-	-	-	-	-	Suggest
Register/edit assets; confirm category	-	-	Asset	-	-	-	-	-	-	Suggest
Propose release or disposition; record execution	-	-	Asset	-	-	-	-	-	-	Suggest
Decide approvals, disputes and Approved cancellation	-	-	-	-	Gate	-	-	-	-	-
Record collection; confirm receipt	-	-	Asset	-	-	-	-	-	-	-

RQ Requester | DR Dept. representative | AC Asset custodian | MS Maintenance staff | WA Workflow administrator
PO Procurement officer | FO Finance officer | SO Sustainability officer | SA System administrator | AI AI services

- Denied; Self own identity/recipient notices; Scope authorized listings; Own own request; Dept department requests (only creator cancels Seeking/Pending); Asset assigned/source/receiving assets; Gate eligible case with no self-approval.

AI describes suggestions, not a profile or database grant. It receives only the task data already authorized for the caller; it cannot execute a business write.

Drivers: FR01-FR18; AI01-AI08; BR03-BR06; NFR03-NFR04, NFR09-NFR10, NFR13; Phase 3 Section 5 permission profiles.

6.2 Access matrix - service, reporting and administration

Default: deny. A scope entry still requires an active account, permitted record, valid action and allowed state.

Action	RQ	DR	AC	MS	WA	PO	FO	SO	SA	AI
Record inspection/service/condition	-	-	Asset	Job	-	-	-	-	-	Suggest
Authorize repair	-	-	Asset	-	-	-	-	-	-	Suggest
View numeric reports; review AI draft	-	-	-	-	Ops	Demand	Finance	Env	-	Draft
Validate reporting inputs	-	-	-	-	-	Demand	Finance	Env	-	-
Manage accounts/profiles/departments	-	-	-	-	-	-	-	-	System	-
Read audit history	-	-	-	-	Ops	-	-	-	System	-
Backup and restore	-	-	-	-	-	-	-	-	System	-

RQ Requester | DR Dept. representative | AC Asset custodian | MS Maintenance staff | WA Workflow administrator
PO Procurement officer | FO Finance officer | SO Sustainability officer | SA System administrator | AI AI services

- Denied; Asset assigned assets; Job assigned service job; Ops operational reports/history; Demand demand and avoided-purchase fields; Finance financial fields; Env environmental/life-extension fields; System account, audit and recovery scope; Draft narrative only.

Custodians/requesters see permitted record history in normal screens. Audit-interface access is limited to reviewers. Evidence and notification links always recheck current parent-record permissions.

Drivers: FR01-FR18; AI01-AI08; BR03-BR06; NFR03-NFR04, NFR09-NFR10, NFR13; Phase 3 Section 5 permission profiles.

Campus Item Exchange | Phase 4 design | Synthetic examples; implementation checks are planned.

6.3 Security controls and enforcement

One central policy and scoped repositories protect every route, file, notification link and AI retrieval.

Control	Design and requirement references
One authorization policy	Use central action/role/scope checks and scoped repositories for protected pages, writes, evidence, notification links and AI retrieval. Recheck active state and current authority on every protected action. FR11, FR13, FR16; BR06; NFR03
Credentials and sessions	Store salted adaptive password hashes. Use HttpOnly session cookies, SameSite=Lax, Secure beyond localhost and CSRF tokens on state-changing forms. Keep AI keys and session secrets out of tracked files, reports, logs and AI payloads. Browser traffic beyond localhost and hosted AI calls use authenticated HTTPS where applicable. NFR04; Phase 3 Section 5 interfaces
Evidence validation	Accept JPEG, PNG and PDF up to 5 MB; reject unauthorized, unsupported or oversized files. Use randomized filenames outside the public web directory; retrieve through an authorized parent-record route; record uploader/time. FR13; NFR03-NFR04
AI isolation	Send only authorized task fields; treat descriptions/documents and returned text as data. Validate IDs, categories, facts, policy sources and report totals. AI has no approval, custody, account or disposal authority. AI07-AI08; NFR03, NFR10; BR04-BR07
Record integrity	Use parameterized SQL. Reservations, handover and disposition use atomic transactions, current-state checks and duplicate-action protection. Append corrections rather than deleting history. Use labelled synthetic data and retain course audit records. FR05, FR08, FR12; BR06; NFR05, NFR10

ADR-11 Authorization enforcement

Options: repeated route-local checks OR central policy plus scoped repositories.
We choose central checks because all nine profiles and AI retrieval must obey the same scope and active-account rules. We give up feature-local policy freedom and accept a shared dependency.
Planned check: test permitted actions and forbidden direct URLs/writes, cross-department data, evidence and AI retrieval for every profile; deactivate an account and retry.

Drivers: FR11-FR13, FR16; AI07-AI08; BR06; NFR03-NFR05, NFR10. Database and evidence remain private; AI has no direct store access.

6.4 Audit records and AI provenance

Append important changes with enough context to reconstruct who changed what and why.

Required event fields

Event ID and linked request/asset/transfer/AI interaction IDs
Actor account, active profile and department
Timestamp with timezone; display campus times in Asia/Dubai
Action, result and reason
Relevant previous and new values
Evidence/input references and review outcome where applicable

Access and corrections

Workflow administrators review relevant workflow history; system administrators review system audit/recovery records. Ordinary users cannot delete audit history. Corrections append linked entries.

AI review record

Retain input references/snapshot, output, time, provider, model/version or Version unavailable, prompt version, confidence or Unavailable, validation outcome and accept/edit/reject/override record. A score is not a correctness probability.

Synthetic event	Example value
Event Id	AUD-0101
Time	2026-10-30T14:20:00+04:00
Actor	U-WA03 (Workflow administrator)
Record	REQ-0101 / MON-014
Action	Reservation approved
Before	Pending; hold MON-014 active
After	Approved; same hold retained; ownership unchanged
Result	Committed

Reason

Eligible serviceable unit and valid destination; proposer U-RQ07 differs from approver.

Coverage

Record important asset, request, approval, receipt, service, disposition, reporting-input and account changes. Record authorization denials and recovery actions as a proposed diagnostic extension without storing credentials or forbidden payloads.

Drivers: FR12; AI07; BR06; NFR04, NFR10. Never log passwords, AI keys, session secrets or unauthorized payloads.

6.5 Backup and recovery

Required snapshot before every demonstration; isolated restoration target: all snapshot records within 10 minutes.

Snapshot schedule and location

Required: a consistent snapshot before each demonstration (NFR09). Proposed extra: also take one at the end of each day in which records change.
Store timestamped sets under a restricted backups directory outside the web-accessible tree; keep a private off-device copy where available. Only the system-administrator recovery procedure may access them. Never place backups or credentials in the public Git repository.
One coordinated set containing the SQLite snapshot, protected evidence files, audit/AI/notification records, schema/runtime-version information and a file/count manifest. Back up password hashes as protected account data; exclude plaintext secrets and provider keys.
Proposed prototype retention: keep demonstration snapshots and the most recent daily snapshots throughout the course, with access restricted. Live-data retention remains subject to institutional review.

Consistency and restore procedure

Pause writes while the application produces the database snapshot and matching evidence-file set; record one snapshot identifier and verify the manifest before resuming writes. Do not copy independently changing database/evidence files and assume they match.
1. Start a timer and select a complete snapshot/manifest; restore to an isolated database and evidence directory, not over the active demonstration data.
2. Check database integrity, record counts, file manifests and referenced evidence; inspect representative assets, holds, approvals, transfers, audit, AI and notification records.
3. Exercise sign-in and a permitted read, verify a forbidden action is denied, and apply normal expiry checks before stock-sensitive actions.
4. Stop the timer; record snapshot ID, tester, environment, expected/actual counts, elapsed time and evidence. Acceptance requires every snapshot record to be restored within 10 minutes; later changes may be lost.

ADR-12 Backup strategy

Options: full coordinated DB/evidence snapshots OR incremental backups with version tracking and replay.
We choose full snapshots because the small prototype makes one inspectable recovery set practical. We give up smaller incremental backups and changes after the selected snapshot; retained full copies use more storage.
Planned check: restore to an isolated database/evidence directory, verify every record, file reference and access check, and record elapsed time against the 10-minute target.

FR12-FR13; BR02, BR06; NFR09-NFR10, NFR12. Restricted off-device copies are a proposed extra, where available. Backups and secrets never enter the public repository.

7.1 Register an asset

Role: Asset custodian - assigned department | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Register an asset

Asset / unit ID: UNIT-M001	Name / short description: 24-inch HDMI monitor
Category: Electronics	Subcategory: Monitors
Quantity: 1	Condition: Serviceable
Owner department: Mechanical Engineering	Custodian: C-101
Location: Lab M-204	Purchase date: 2024-09-01
Estimated value: AED 300.00	Evidence: front.png - 1.2 MB

Save draft | Request release

Classification suggestion

Suggested fields

Electronics / Monitors; tags: display, HDMI. Material: Unknown.

Confidence

Unavailable - review required.

Reason and source

The title and front.png show a monitor. Unseen ports and power use remain Unknown.

Your decision

Review or edit the fields before saving. A draft is not published.

Review / edit | Ignore

If AI is unavailable

Keep all entered fields. Select the category manually and save the draft; release still needs workflow-administrator approval.

Decision: The custodian reviews classification and saves a draft or proposes release.

Phase 3: FR01-FR02, FR13, FR17; AI02, AI07-AI08; UC02, UC10

Screen 1 of 8

7.2 Submit a resource request

Role: Requester - own request | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Submit a resource request

Purpose: Teaching laboratory	Category / specifications: Monitor / 24-inch HDMI
Quantity / preferred condition: 1 / Serviceable	Urgency: Normal
Needed date: 2026-11-11	Destination department: Chemistry
Receiving custodian: C-202	Destination location: Lab C-107
Selected unit: UNIT-M001	Current requestable quantity: 1 - checked again on confirm

Application rules

Confirm creates Pending and a stock hold. Seeking saves a need without reserving stock.

Confirm request | **Save as Seeking**

Assisted request preview

Proposed need

One 24-inch HDMI monitor for the Chemistry teaching laboratory by 11 November.

Confidence

Unavailable - review required.

Reason and source

The preview uses your purpose, quantity, specifications and destination fields.

Your decision

Review every field. No request or stock hold exists until you confirm.

If AI is unavailable

Keep the request fields and selected unit. Use keyword filters or enter the need manually; normal stock and permission checks still apply.

Decision: The requester confirms the exact request, or saves an unmatched need as Seeking.

Phase 3: FR03-FR06; AI01, AI04, AI08; BR01-BR03; UC03, UC12

7.3 Review recommendations

Role: Requester or department representative - permitted listings | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Review recommendations

Need: 1 serviceable HDMI monitor; Chemistry; due 2026-11-11

All shown units passed permission, publication and safety checks.

Eligible unit	Compatibility	Known factors
○ UNIT-M001	92 / 100	HDMI; cost AED 20.00; no repair
○ UNIT-M003	86 / 100	HDMI; cost AED 25.00; no repair
○ UNIT-M011	72 / 100 - weak	HDMI; repair cost Unknown

Scores describe compatibility, not correctness probability.

Top-two gap: 6 points. No default or emphasized top choice.

Continue with selection | Manual search

Why these matches

Illustrative factor contributions

UNIT-M001: suitability 50 + urgency 10 + transfer 12 + repair 10 + diversion 5 + carbon 5 = 92 points.

Confidence

Unavailable - review required.

Reason and source

UNIT-M001 meets the HDMI need. Its known transfer cost is lower than UNIT-M003.

Uncertainty

A small score gap and missing inputs need review. Unknown facts are not zero.

If AI is unavailable

Show current keyword/filter results in rule order, labelled Manual search results. Preserve the need; do not invent AI scores or explanations.

Decision: The requester chooses explicitly; submission rechecks stock before making a hold.

Phase 3: FR03-FR05; AI01, AI06-AI08; BR01-BR02; UC09

7.4 Approve a transfer request

Role: Workflow administrator - no self-approval | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Approve a transfer request

Request / requester	Reviewer
REQ-014 / R-201	WA-01 - different from requester
Unit / quantity	Reservation
UNIT-M001 / 1	Pending - hold active
Source department / custodian	Destination / custodian
Mechanical / C-101	Chemistry / C-202
Destination location	Safety / publication
Lab C-107	Serviceable / Published
Hold expiry - Asia/Dubai	Decision reason
2026-11-01 14:00	Meets the recorded teaching need

Application rules

Approval retains the hold. Collection evidence and receiving-custodian receipt are required before custody changes.

Approve with reason | **Reject with reason**

Saved match explanation

Existing request evidence

Illustrated match score: 92 / 100 for UNIT-M001. Read-only explanation from the request.

Confidence

Unavailable - review required.

Reason and source

The recorded HDMI requirement and available quantity match the asset record.

Authority

AI cannot approve. Recheck role, requester, stock, safety and expiry before saving.

If AI is unavailable

Decide using the request, asset facts and evidence. New AI output is unnecessary; the server still enforces every approval rule.

Decision: The administrator approves or rejects with a reason; receipt remains a separate human gate.

Phase 3: FR07-FR08, FR12, FR15, FR17; AI08; BR02-BR06; UC05-UC06

Screen 4 of 8

7.5 Track maintenance

Role: Maintenance staff - assigned service job | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Track maintenance

Job / assigned staff	Asset
JOB-008 / MS-011	UNIT-M007 - HDMI monitor
Reported defect	Serviceability
Intermittent display flicker	Under service - allocation blocked
Inspection date	Recorded finding
2026-10-30	Power board fault
Service state	Repair estimate
Inspected	AED 50.00
Evidence	Repair authority
inspection_008.pdf - 0.6 MB	Custodian C-101 - pending decision

Application rules

Saving an inspection does not clear safety. Repair authorization and recorded human clearance remain required.

Save inspection | **Request repair review**

Saved custodian advice

Suggested comparison

Compare repair at AED 50.00 with evidenced replacement at AED 300.00.

Confidence

Unavailable - review required.

Reason and missing inputs

A repair may extend service. Carbon factors and revised retirement date are unavailable.

Authority

Advice is read-only here. Staff record findings; the custodian authorizes repair. AI cannot clear safety.

If AI is unavailable

Continue the inspection and service form manually. Preserve findings and evidence; the asset remains blocked until proper human clearance.

Decision: Assigned staff record the inspection; the custodian reviews repair authorization.

Phase 3: FR09-FR10, FR13-FR14, FR17; AI03, AI08; NFR13; UC07, UC11

7.6 Review sustainability dashboard

Role: Sustainability officer - environmental and life-extension scope | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Review sustainability dashboard

Period: 2026-10-01 to 2026-10-30 | Evidenced completed activity

Stock and open needs **8 available; 3 open needs**	Reuse - M02 **4 units; 4 / 20 = 20%**
Waste diversion - M03 **3 / 4 = 75%**	Estimated carbon - M04 **40 kg CO2e; coverage 2 / 4**
Estimated extension - M05 **120 days / 2 assessed units**	Observed extra service **60 days; separately recorded**

Application rules

Count distinct, evidenced units. Sample carbon factors: 25 - 5 = 20 kg CO2e per covered unit; missing factors stay Unavailable.

Review environmental inputs | **Request summary**

Activity summary draft

Illustrated AI result

Four units were reused. Three of four final outcomes were diverted from disposal.

Confidence

Unavailable - review required.

Reason and source

The draft uses application totals M02-M04. The 40 kg estimate covers only 2 of 4 reused units.

Your decision

Check environmental evidence and coverage. AI cannot create missing factors or count planned actions.

If AI is unavailable

Keep the calculated indicators, source links and coverage visible. Show Summary unavailable; environmental input review still works.

Decision: The sustainability officer validates assigned evidence and reviews the narrative against totals.

Phase 3: FR18; AI05, AI07-AI08; BR07; M02-M05; NFR11; UC13

Screen 6 of 8

7.7 Generate a financial report

Role: Finance officer - financial input and report scope | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Generate a financial report

Reporting period: 2026-10-01 to 2026-10-30

Report type: Avoided purchases and net savings

Replacement cases and source costs - AED

Case	Avoided value	Transfer	Repair / refurb.	Net savings
TR-006	300.00	20.00	30.00 / 0.00	250.00
TR-007	200.00	Unknown	0.00 / 0.00	Unavailable

Known-case subtotal: AED 250.00 | Coverage: 1 of 2 cases

Complete net total: Unavailable - TR-007 has an unknown cost.

Validate financial inputs | Request narrative

Financial narrative draft

Illustrated AI result

Known net savings are AED 250.00 for 1 of 2 cases. The complete total is Unavailable.

Confidence

Unavailable - review required.

Reason and source

TR-006: 300 - 20 - 30 - 0 = 250. TR-007 lacks an evidenced transfer cost.

Your decision

Review figures, period, case links and assumptions. Reject any invented or mismatched amount.

Use reviewed draft

If AI is unavailable

The numeric report remains available. Show Draft unavailable and retain period and criteria; no unchecked narrative is presented.

Decision: The finance officer validates financial inputs and reviews the draft against the numeric report.

Phase 3: FR12, FR18; AI05, AI07-AI08; BR07; M01; NFR11; UC13

7.8 Use the conversational assistant

Role: Requester - authorized catalogue and own requests | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Use the conversational assistant

You

Find one HDMI monitor for Chemistry by 11 November.

Assistant

UNIT-M001 fits the recorded need. Source: catalogue UNIT-M001. No approved campus collection policy is available.

Exact request preview - no hold yet

UNIT-M001; quantity 1; purpose Teaching laboratory; 24-inch HDMI; Serviceable; Normal urgency; needed 2026-11-11.

Destination: Chemistry; custodian C-202; Lab C-107.

Confirm request | Edit preview | Cancel

Facts, suggestion and control

Confidence

Unavailable - review required.

Reason and source

The authorized asset record lists HDMI, serviceability and one requestable unit.

Application rules

Recheck permission, current stock, dates and destination when you confirm. The assistant cannot reserve stock.

Your decision

Edit or cancel the preview, or explicitly confirm it through the ordinary request validation.

If AI is unavailable

Retain the question and preview. Show Assistant unavailable and provide the ordinary catalogue, status and request screens.

Decision: The requester confirms the exact preview; only normal validation can create a request and hold.

Phase 3: FR03-FR06, FR16; AI04, AI07-AI08; BR01-BR02; UC12

Screen 8 of 8

AI-use and review statement

Tool use is disclosed separately from the team review and implementation acceptance still required.

Item	Disclosure
Tool	OpenAI Codex.
Drafting	Assisted with architecture choices, AI service policies, approval points, access rules, recovery procedures and decision-record wording, using the supplied Phase 3 requirements and Phase 4 lecture.
Visuals	Generated vector context/DFD and architecture figures, tables and eight synthetic screen wireframes. Reference images guided the blue styling, sparse diagram detail and Level 2 dotted boundary.
Document checks performed	Compared context and Level 1 external flow names/directions/entities and the six Level 2 boundary flows; reviewed six process/five store mappings, all nine profiles plus AI, stated deadlines and example arithmetic; rendered and visually checked pages. These are checks of the design document.
Human responsibility	Each member must verify the design and requirement interpretation, review AI-assisted content, and commit their own contribution. Provider access, capabilities, institutional rules and all planned implementation/acceptance checks still require actual evidence.

Source documents

Campus Item Exchange Phase Three Requirements Document, 4 October 2026 (Phase_3_04_Cumulative.docx).
Phase 4 Architecture Design, Fall 2026 (Phase_4_Architecture_Design_fall26_Blackboard.pptx), especially slides 3-4 and 15-24.

AI suggestions never replace the nine-profile permission matrix, required human approval, recorded safety clearance or evidence-based reporting.